Ibexolimod Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-767/A

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The control strategy links critical quality attributes to critical process parameters identified during development. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. No changes to the approved analytical procedures are proposed in this submission. The control strategy links critical quality attributes to critical process parameters identified during development. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

The control strategy links critical quality attributes to critical process parameters identified during development. No changes to the approved analytical procedures are proposed in this submission. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

P.4 Control of Excipients

For the commercial formulation F-767/A, Ac-Di-Sol SD-711 (Croscarmellose sodium, disintegrant) is used.

The approach described herein is consistent with Ph. Eur. general monograph 2034.

Selection Justification

The selection of Ac-Di-Sol SD-711 is justified by reduced lot-to-lot variability reported by the qualified supplier observed during development studies.

Current Commercial Specification

The current approved commercial specification (SPEC-EX-1590 Rev 02) lists Croscarmellose sodium as Solutab Type A.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

No changes to the approved analytical procedures are proposed in this submission. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.